

JONGHA (JON) RYU

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RESEARCH INTERESTS	Mathematical and statistical foundations of scientific machine learning, with recent focus on neural spectral methods [C8], [C14], [C15], probabilistic and generative modeling [C11], [C12], [P5], and uncertainty quantification and sequential decision making with anytime-valid inference [J2], [C9], [C10], [C13].
EMPLOYMENT	Massachusetts Institute of Technology (MIT) Aug. 2022 – present Postdoctoral Associate, Department of EECS/Research Laboratory of Electronics · Advisor: Prof. Gregory W. Wornell.
EDUCATION	University of California San Diego (UCSD) Sep. 2015 – Jun. 2022 Ph.D. in Electrical Engineering (GPA: 3.99/4.0) · Thesis advisors: Prof. Young-Han Kim and Prof. Sanjoy Dasgupta. · Thesis title: “From Information Theory to Machine Learning Algorithms: A Few Vignettes”. M.S. in Electrical Engineering Dec. 2018 Seoul National University (SNU) Mar. 2008 – Aug. 2015 Bachelor of Science (<i>summa cum laude</i> , GPA: 4.11/4.3) · Double major in Electrical and Computer Engineering and Mathematical Sciences ; minor in Physics .
JOURNAL PAPERS	(* indicates equal contribution. † indicates alphabetical order.) [J1] J. Jon Ryu* , S. Ganguly*, Y.-H. Kim, Y.-K. Noh, D. Lee, “Nearest neighbor density functional estimation from inverse Laplace transform,” <i>IEEE Trans. Inf. Theory</i> , vol. 68, no. 6, pp. 3511–3551, Jun. 2022. [J2] J. Jon Ryu , A. Bhatt, “On confidence sequences for bounded random processes via universal gambling strategies,” <i>IEEE Trans. Inf. Theory</i> , vol. 70, no. 10, pp. 7143–7161, Oct. 2024.
CONFERENCE PAPERS	[C1] Alankrita Bhatt†, Jiun-Ting Huang†, Young-Han Kim†, J. Jon Ryu† , Pinar Sen†, “Monte Carlo methods for randomized likelihood decoding,” <i>Annual Allerton Conference on Communication, Control, and Computation (Allerton)</i> , Sep. 2018. [C2] Jongha Ryu , Young-Han Kim, “Conditional distribution learning using neural networks and its application to universal image denoising,” <i>IEEE International Conference on Image Processing (ICIP)</i> , Oct. 2018. [C3] Alankrita Bhatt†, Jiun-Ting Huang†, Young-Han Kim†, J. Jon Ryu† , Pinar Sen†, “Variations on a theme by Liu, Cuff, and Verdú: The power of posterior sampling,” <i>IEEE Information Theory Workshop (ITW)</i> , Nov. 2018. [C4] Yang Yang, Guillaume Sautiere, J. Jon Ryu , Taco Cohen, “Feedback Recurrent Autoencoder,” <i>IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)</i> , May 2020. [C5] J. Jon Ryu , Jiun-Ting Huang, Young-Han Kim, “On the Role of Eigendecomposition in Kernel Embedding,” <i>IEEE International Symposium on Information Theory (ISIT)</i> , Jun. 2021. [C6] J. Jon Ryu , Alankrita Bhatt, Young-Han Kim, “Parameter-free Online Linear Optimization with Side Information via Universal Coin Betting,” <i>International Conference on Artificial Intelligence and Statistics (AISTATS)</i> , Apr. 2022. [C7] Alankrita Bhatt*, J. Jon Ryu* , Young-Han Kim, “On Universal Portfolios with Continuous Side Information,” <i>AISTATS</i> , Apr. 2023. [C8] J. Jon Ryu , X. Xu, H. S. M. Erol, Y. Bu, L. Zheng, G. Wornell, “Operator SVD with neural networks via nested low-rank approximation,” <i>International Conference on Machine Learning (ICML)</i> , Jul. 2024.

- [C9] **J. Jon Ryu**, G. W. Wornell, “Gambling-based confidence sequences for bounded random vectors,” in *ICML*, Jul. 2024. **Spotlight** (top 3.5%).
- [C10] M. Shen*, **J. Jon Ryu***, S. Ghosh, Y. Bu, P. Sattigeri, S. Das, G. W. Wornell, “Are uncertainty quantification capabilities of evidential deep learning a mirage?,” in *Annual Conference on Neural Information Processing Systems (NeurIPS)*, Dec. 2024.
- [C11] T. Jayashankar*, **J. Jon Ryu***, G. W. Wornell, “Score-of-mixture training: Training one-step generative models made simple via score estimation of mixture distributions,” in *ICML*, Jul. 2025. **Spotlight** (top 2.6%).
- [C12] **J. Jon Ryu**, A. Shah, G. W. Wornell, “A unified view on learning unnormalized distributions via noise-contrastive estimation,” in *ICML*, Jul. 2025.
- [C13] **J. Jon Ryu**, J. Kwon, B. Koppe, K.-S. Jun, “Improved offline contextual bandits with second-order bounds: Betting and freezing,” in *Annual Conference on Learning Theory (COLT)*, Jun. 2025.
- [C14] M. Jeong*, **J. Jon Ryu***, S.-Y. Yun, G. W. Wornell, “Efficient parametric SVD of Koopman operator for stochastic dynamical systems,” in *NeurIPS*, Dec. 2025.
- [C15] **J. Jon Ryu**, S. Zhou, G. W. Wornell, “Revisiting orbital minimization for neural operator decomposition,” in *NeurIPS*, Dec. 2025.

PREPRINTS

- [P1] **J. Jon Ryu**, Y. Choi, Y.-H. Kim, M. El-Khamy, J. Lee, “Learning with succinct common representation with Wyner’s common information,” arXiv:1905.10945. (Preliminary versions presented at *NeurIPS Bayesian Deep Learning Workshops (2018, 2021)*).
- [P2] **J. Jon Ryu**, Young-Han Kim, “An Information-Theoretic Proof of Kac–Bernstein Theorem”, arXiv:2202.06005.
- [P3] **J. Jon Ryu**, Young-Han Kim, “Minimax Optimal Algorithms with Fixed- k -Nearest Neighbors”, arXiv:2202.02464v3.
- [P4] Tejas Jayashankar*, **J. Jon Ryu***, Xiangxiang Xu, Gregory W. Wornell, “Lifted Residual Score Estimation”. (A preliminary version was presented at *ICML 2024 Workshop on Structured Probabilistic Inference & Generative Modeling*.)
- [P5] **J. Jon Ryu**, P. Yeddanapudi, X. Xu, G. W. Wornell, “Contrastive predictive coding done right for mutual information estimation,” arXiv:2510.25983.

INVITED TALKS

- **From Information Theory to Machine Learning Algorithms: Two Vignettes**, Signals, Information and Algorithms Laboratory, MIT, 2022; Center for AI and Natural Sciences, KIAS, 2022.
- **From Wyner’s Common Information to Learning with Succinct Representation**, Information Theory and Applications Workshop (ITA), 2022; Machine Intelligence and Data Science Laboratory, SNU, 2023; Inference and Information for Data Science Lab, KAIST, 2023.
- **Nearest Neighbor Density Functional Estimation from Inverse Laplace Transform**, Center for AI and Natural Sciences, KIAS, 2022.
- **On Confidence Sequences from Universal Gambling**, Prof. Aaditya Ramdas’s Group, CMU, 2022; Hanyang University, 2023.
- **Operator SVD with Neural Networks via Nested Low-Rank Approximation**, MLTea, MIT, 2023; ITA Workshop, 2024; MERL, 2024; KAIST AI, 2024; KIAS, 2024; Flatiron Institute, 2024; MIT JTL Urban Mobility Lab, 2025; Prof. Devavrat Shah’s Group, MIT, 2025.
- **Efficient Generative Modeling and Operator Learning from First Principles**, Frontier Research, Prescient Design, Genentech, 2025; Department of Electrical Engineering, POSTECH, 2025; Department of Computer Science and Engineering, POSTECH, 2025.
- **Tools for Scalable and Reliable Scientific Inference from First Principles**, Prescient Design, Genentech, 2025.

HONORS AND AWARDS	Spotlight Recognitions (top ~3%)	ICML 2024, ICML 2025
	Top Reviewer Awards	NeurIPS 2023, NeurIPS 2025
	Departmental Fellowship Department of ECE, UCSD	Sep. 2015 – Jun. 2016
	Kwanjeong Scholarship for Graduate Study Kwanjeong Scholarship Foundation, South Korea	Sep. 2015 – Jun. 2020
	Kwanjeong Scholarship for Undergraduate Study Kwanjeong Scholarship Foundation, South Korea	Mar. 2010 – Dec. 2013
	University Students Contest of Mathematics Korean Mathematical Society · Among non-math majors: Gold prize (2010), honorable mention (2009). · Among math majors: Bronze prize (2013).	
PROFESSIONAL SERVICE	• Journals: IEEE Trans. Inf. Theory, IEEE J. Sel. Areas Inf. Theory. • Conferences: ISIT (2017, 2023, 2024, 2025), ITW (2022), AISTATS (2022, 2023, 2024), ICML (2022–2025), NeurIPS (2022–2025, top reviewer 2023 and 2025), ICLR (2024, 2026).	
INTERNSHIP EXPERIENCE	Research Intern AI Research Group, Qualcomm Technologies, Inc. · Researched deep-learning-based compression for speech processing [C4].	Jun. 2019 – Dec. 2019
	Research Intern Deep Learning Team, SoC R&D, Samsung Semiconductor Inc. · Developed an information-theoretic representation learning principle [P1].	Jun. 2018 – Sep. 2018
TEACHING EXPERIENCE	Instructor (MIT)	
	• 6.7800 Inference and Information · Designed and taught a five-lecture sequence of advanced special sessions as co-instructor. · Topics: minimax bit prediction, group testing, universal inference and concentration, universal learning approach for Rock–Paper–Scissors machine, variational perspective on generative modeling. · Evaluations: overall rating 7.0/7 (Spring 2024), 6.3/7 (Spring 2025).	Spring 2024, Spring 2025
	Teaching Assistant (UCSD)	
	• ECE 250 Random Processes	Winter 2017
	• ECE 154C Communication Systems · Designed hands-on programming assignments using Julia. · Topics: basic source coding and channel coding algorithms.	Spring 2017
	• ECE 225B Universal Probability and Applications in Data Science · Designed hands-on programming assignments using Python. · Topics: Lempel–Ziv probability assignment, context-tree weighting, universal portfolio. • ECE 269 Linear Algebra and Applications	Spring 2018 Winter 2019

MENTORING EXPERIENCE	Tejas Jayashankar , Ph.D. student, MIT (now at Meta) Sep. 2022 – Jun. 2025 · Worked on generative modeling and representation learning; co-authored a spotlight paper on one-step image generation (ICML 2025) [C11].		
	Abhin Shah , Ph.D. student, MIT (now at Five Rings) Sep. 2022 – Aug. 2024 · Worked on fairness and unnormalized models; co-authored papers on fairness (ISIT 2024) and noise-contrastive estimation (ICML 2025) [C12].		
	Maohao Shen , Ph.D. student, MIT Sep. 2022 – present · Worked on evidential deep learning; co-authored a paper characterizing its behavior (NeurIPS 2024) [C10]; working on RLHF and memory-based LLM methods.		
	Minchan Jeong , Ph.D. student, KAIST Sep. 2024 – present · Worked on applications of neural spectral methods; co-authored a paper on stochastic dynamical systems (NeurIPS 2025) [C14]; working on excited-states computation in quantum chemistry.		
	Rachana Madhukara , Ph.D. student, MIT Mar. 2025 – present · Working on Koopman-operator-based approaches for efficient generative modeling.		
	Axel Adjei , M.Eng. student, MIT Sep. 2025 – present · Working on theoretical analysis of convergence behavior of iterative spectral decomposition algorithms; developing guarantees for a new streaming canonical correlation analysis algorithm.		
	Samuel Zhou , undergraduate student, MIT Mar. 2025 – present · Worked on neural spectral methods; co-authored a paper on linear-algebraic operator decomposition (NeurIPS 2025) [C15]; working on interpretable representation learning in LLMs.		
	Pavan Yeddanapudi , undergraduate student, MIT Sep. 2024 – present · Worked on information-measure estimation; co-authored a paper on mutual information estimation (arXiv 2025, under review) [P5]; working on score-based representation learning.		
OTHER EXPERIENCE	Military service (mandatory) Mar. 2011 – Dec. 2012 Republic of Korea Army		
REFERENCES	Gregory W. Wornell Professor of EECS, MIT Postdoc advisor ✉ gww@mit.edu	Young-Han Kim Professor of ECE, UCSD Ph.D. advisor ✉ yhk@ucsd.edu	Sanjoy Dasgupta Professor of CSE, UCSD Ph.D. advisor ✉ sadasgupta@ucsd.edu